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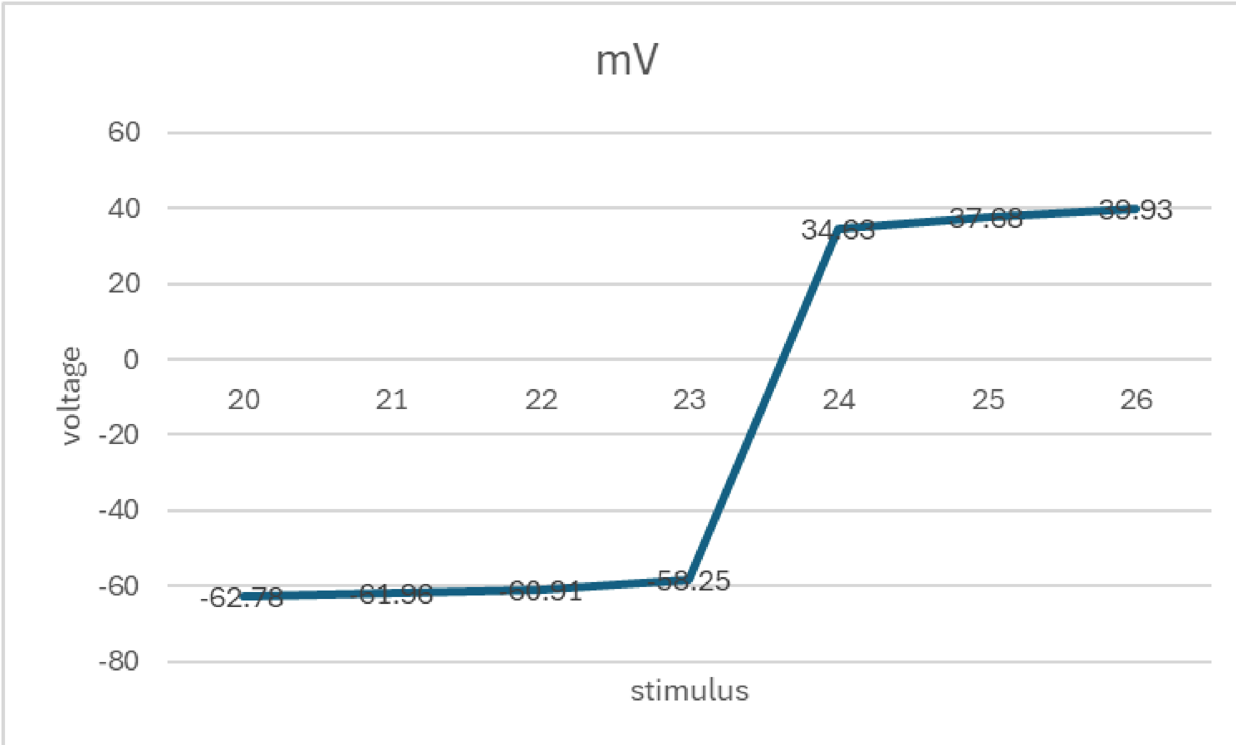
NSc 4130 Lab
2/10/20206

NIA lab 2: Neuromembrane

1. We did not get an action potential, no
2. The highest voltage was -62.77 mV
3. No, still no action potential
4. Highest voltage was -61.96 mV

Simulation #	Stimulus intensity	mV	RMP	Absolute difference
1	20	-62.78	-70.38	7.6
2	21	-61.96	-70.38	8.42
3	22	-60.91	-70.38	9.47
4	23	-58.250	-70.38	12.13
5	24	34.63	-70.38	105.31
6	25	37.68	-70.38	108.03
7	26	39.93	-70.38	110.31

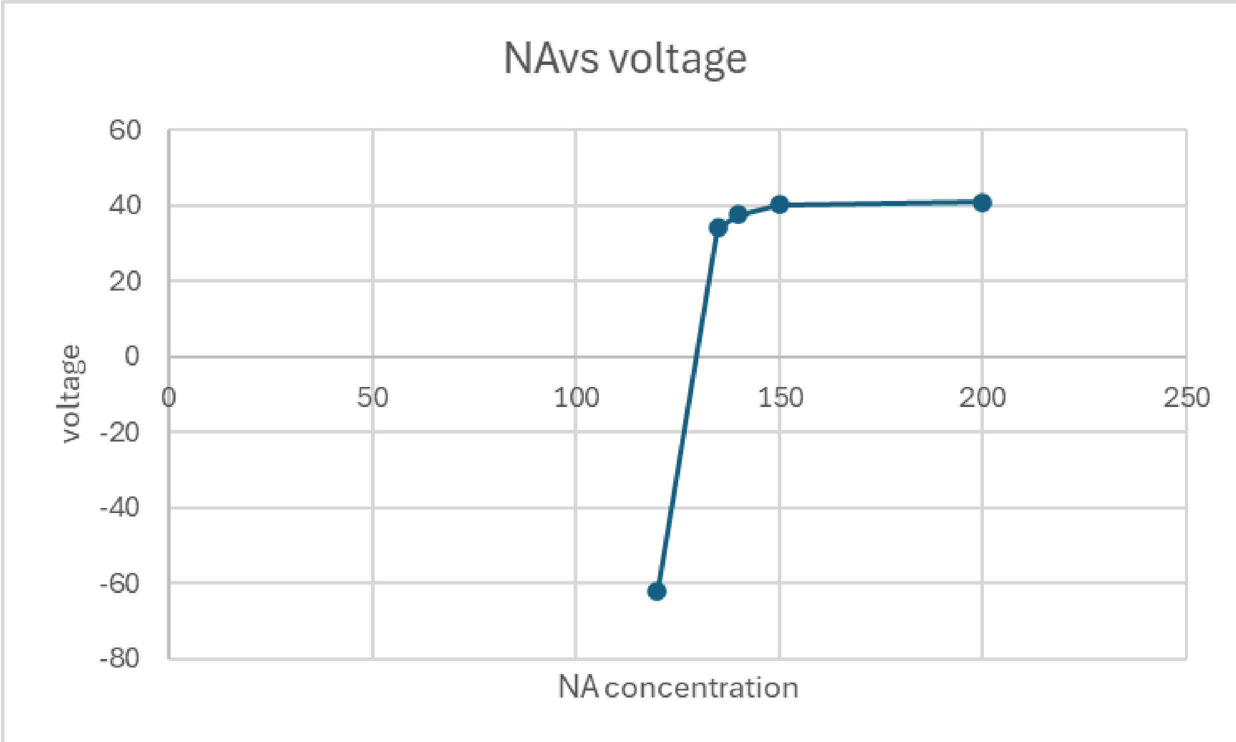
graph , x is stimulus intensity, and y is peak voltage



We can see the all or nothing principle, because until the threshold was reached, we never saw an AP, even if it was just barely lower.

simulation#	Na	Time to max	Max voltage	RMP	Absolute diff
1	120	12	-62.2	-72.144	9.94
2	135	12.5	34.1	-70.78	104.88
3	140	12.2	37.68	-70.38	108.06
4	150	12	40.28	-69.6	109.88
5	200	11.4	40.91	-66.6	107.71

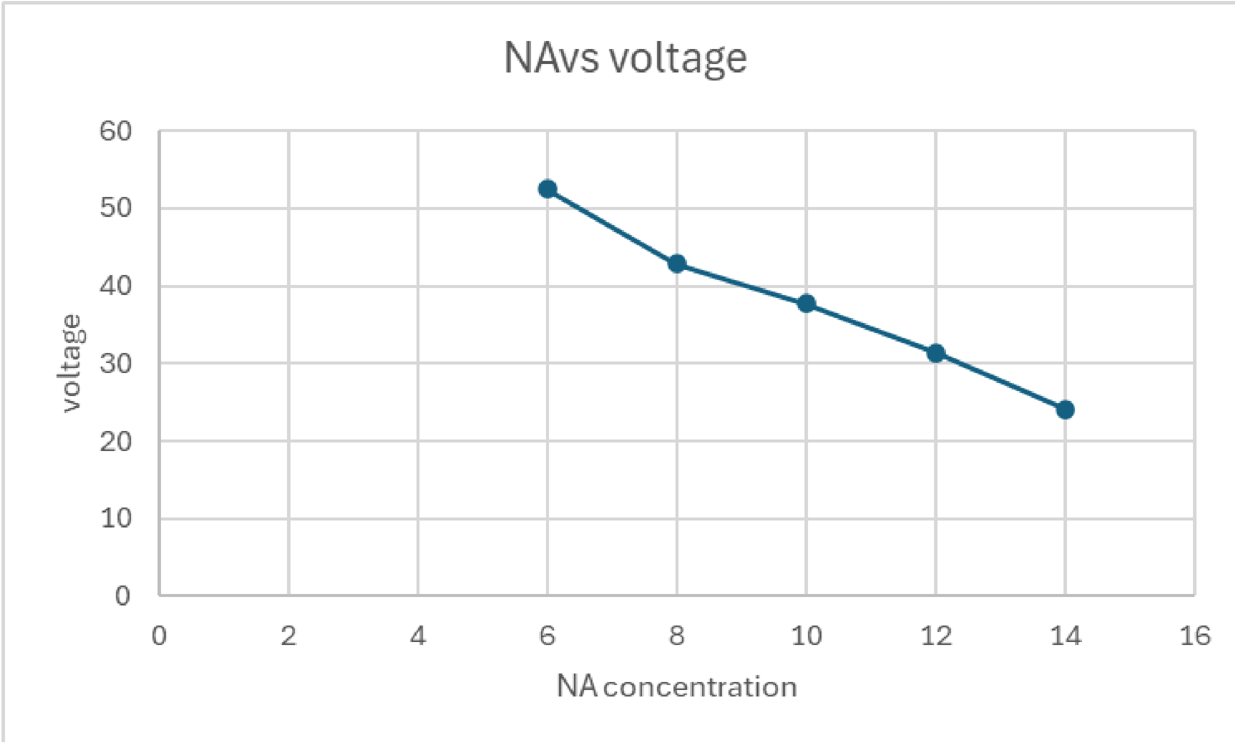
The chart:



8. It looks like there is a certain threshold of concentration outside the cell that is needed in order to have an action potential. Around 135. The higher the concentration outside the cell, the faster the speed for AP. this happens because there's more difference in the electrostatic pressure, so the NA enters the cell faster.

simulation	NA	Time to max	Max voltage	RMP	Absolute difference
1	6	11.9	52.5	-70.32	122.82
2	8	12	42.82	-70.36	113.18
3	10	12.2	37.68	-70.38	108.06
4	12	12.4	31.35	-70.4	101.75
5	14	12.8	24.1	-70.42	94.52

Chart of voltage vs na concentration inside the cell



7.the higher the concentration of NA inside the cell, the lower the AP, because there is less electrostatic pressure comparatively to outside the cell. It also linearly increases the AP speed, as the lower pressure means NA moves slower. Also, the RMP very slightly becomes a bit more negative, but its very very small.

simulation	K	RMP	Time to max	Max voltage	Time to min	Min voltage	AP duration
1	4	-70.38	12.2	37.68	13.3	-87.18	1.1
2	5	-68.35	11.6	34.71	12.8	-82.35	1.2
3	6	-66.4	11.4	33.68	12.8	-78.25	1.4
4	7	-66.4	11.3	28.59	12.5	-74.95	1.2
5	8	-62.5	11.2	21.37	12.6	-71.76	1.4

a) RMP: becomes less negative
Why: less K^+ leaves the cell.

b) Max voltage (AP height): decreases.
Why: more Na^+ channels are inactivated + less Na^+ driving force.

c) Time to max: slightly decreases
Why: starts closer to threshold.

d) Min voltage becomes less negative
Why: K^+ equilibrium potential is less negative, so you can't hyperpolarize as much.

e) AP duration: slightly increases / tends longer.
Why: weaker K^+-driven repolarization.